

HERA-PdM: A Reference Architecture for Uncertainty-Gated, Human-Approved Predictive Maintenance in Industry 5.0

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Abstract: This design paper proposes HERA-PdM, a reference architecture for hierarchical predictive maintenance (PdM) in human-centric Industry 5.0 plants. Four software agents (edge monitoring, Digital Twin state, fog/cloud prognostics and maintenance decision) exchange explicit messages; every remaining-useful-life (RUL) estimate carries a conformally calibrated interval; the edge escalates to higher tiers only when its interval cannot fix a maintenance action; and schedule/urgent actions are issued as explained recommendations that require human approval. We specify the agents, message schemas, decision rules and the safety property the escalation rule provides, describe how the architecture maps onto a production line through an explicitly hypothetical deployment scenario, and summarize preliminary evidence from a companion empirical study on two public bearing datasets, which shows when the design is expected to help and when it is not. We close with the open problems that a deployment would have to solve, notably bearing-level alarm-timing guarantees and operator workload.

Index Terms: predictive maintenance, reference architecture, conformal prediction, edge computing, Digital Twin, human-in-the-loop, Industry 5.0

I. MOTIVATION AND DESIGN GOALS

Predictive maintenance aims to intervene before failure using condition data [1]. In plants moving towards Industry 5.0, three constraints shape any PdM system: sensing happens at the machine and bandwidth to central servers is limited [2]; predictions must state how far they can be trusted [3], [4]; and humans remain responsible for consequential actions [5], while decision-makers often do not adopt opaque advice [6]. Hierarchical and agentic PdM systems exist [7]–[9], but they escalate on anomaly thresholds or uncalibrated confidence [7], [10]. HERA-PdM is designed around five goals:

- G1 *Local by default:* routine monitoring runs on the machine.
- G2 *Calibrated uncertainty everywhere:* every RUL estimate is an interval with a stated coverage level [11], [12].
- G3 *Decision-driven escalation:* computation moves up only when the local estimate cannot decide the maintenance action.
- G4 *Human authority:* high-impact actions are recommendations with evidence, never automatic.
- G5 *Auditability:* all numbers in a recommendation are traceable to a validated predictor; no generative model produces quantitative content.

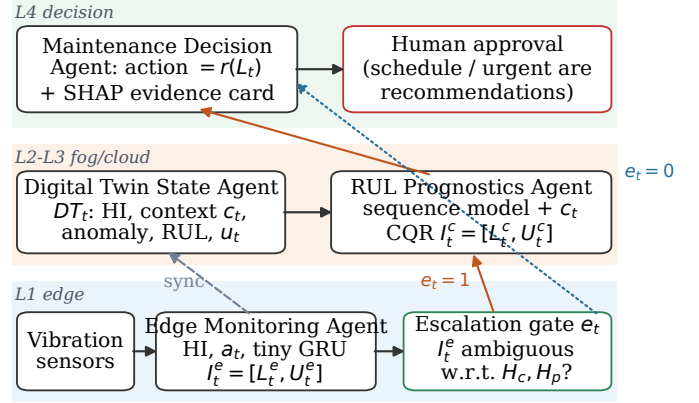


Fig. 1. HERA-PdM reference architecture: agents, tiers and messages.

TABLE I
AGENTS AND MESSAGE INTERFACES.

Agent (tier)	Responsibility	Message out
Edge Monitoring (L1)	features x_t , health index h_t , anomaly a_t , tiny quantile model, calibrated I_t^e , gate e_t	sync (t, h_t, a_t) each window on
Digital Twin State (L2)	synchronized record DT_t , degradation context c_t , sequence model on window $+ c_t$, calibrated I_t^c	to prognostics; state history $(\hat{R}_t, I_t^c)$
RUL Prognostics (L3)	action $r(L_t)$, SHAP evidence, card	recommendation; approval request
Maintenance Decision (L4)	approve / reject / defer	decision log (audit)
Human operator		

II. ARCHITECTURE

Fig. 1 shows the tiers. Table I lists each agent's inputs, outputs and message.

Uncertainty layer. Each tier outputs quantiles that are conformalized; with cross-fitting (CV+ [12]) all historical assets serve as calibration data. The action of an interval is $r(L_t)$ with regions URGENT ($\leq H_c$), SCHEDULE ($\leq H_p$) and CONTINUE.

Escalation rule. The edge escalates when $I_t^e = [L_t^e, U_t^e]$ straddles a boundary in $\{H_c, H_p\}$ or when anomaly evidence contradicts a CONTINUE verdict. A locally resolved CONTINUE requires $L_t^e > H_p$, so a critical asset is left running at the edge only when the true RUL lies outside the calibrated

interval:

$$\Pr(e_t=0, \pi_t=\text{CONTINUE}, y_t \leq H_c) \leq \Pr(y_t \notin I_t^e). \quad (1)$$

Human-in-the-loop. The decision agent emits a card with predicted RUL, interval, anomaly evidence, top SHAP [13] contributors and a templated rationale; SCHEDULE/URGENT wait for approval, and approvals are logged.

III. ILLUSTRATIVE DEPLOYMENT SCENARIO (HYPOTHETICAL)

This section is a design walk-through. All quantities in it are assumptions chosen for illustration; none is a measurement.

Consider a hypothetical packaging line with conveyor drive bearings instrumented with two-axis accelerometers. An edge node per machine extracts features each snapshot and keeps only the healthy baseline and a short window. While intervals stay inside the CONTINUE region the node sends only the 12-byte sync message, which the fog-level Digital Twin uses to maintain long-horizon context. When a bearing's edge interval begins to straddle H_p , the node streams its feature window to the fog prognostics agent, whose interval and SHAP evidence populate a card routed to the shift maintenance planner. The planner approves a scheduled replacement for the next planned stop; an urgent recommendation would additionally notify the line supervisor. In such a deployment, H_c and H_p would be set from the plant's spare-part lead time and planned-stop calendar, and the calibration set would be refreshed as new run-to-failure histories accumulate.

IV. PRELIMINARY EVIDENCE (MEASURED)

All numbers in this section come from the companion empirical study (public data, bearing-grouped cross-validation, three seeds). On FEMTO (17 bearings, abrupt failures) and XJTU-SY (15 bearings, gradual degradation) [14], [15]: cross-fitted CQR reached 0.94–0.96 and 0.93–0.99 coverage at nominal 0.90; calibrated decisions left 0.6% and 9.2% of critical snapshots without action versus 49.5% and 34.2% for point estimates with fixed thresholds. On XJTU-SY the escalation rule matched always-cloud safety with less false maintenance (53.2% vs. 69.1%) while escalating 79.1% of snapshots; on FEMTO it escalated 97.1% and gave no benefit. The decision-sufficiency rule behaved like an interval-width gate, and per-snapshot calibration produced premature bearing-level alarms.

V. OPEN PROBLEMS

Alarm-timing guarantees. Conformal risk control [16] can bound the probability that a bearing's alarm is late, but with about a dozen calibration bearings it met the target only by alarming early (94.1% and 80.0% premature); fleet-scale run-to-failure data are needed. *Operator workload.* Wide intervals shift burden to humans (approval requests on 55.8% of snapshots on XJTU-SY); workload-aware recommendation policies and user studies are required. *Degradation observability.* When failures give no vibration warning, no routing rule helps; richer sensing is a prerequisite. *Embedded validation.* Edge inference was timed on a host CPU (0.15 ms); measurements on MCU/SBC hardware are pending.

VI. CONCLUSION

HERA-PdM specifies how calibrated uncertainty can drive both computation placement and human-approved maintenance recommendations in a hierarchical plant architecture. Preliminary evidence indicates that the design is worthwhile when degradation is observable early and that alarm-timing guarantees, operator workload and embedded validation are the next problems to solve.

REFERENCES

- [1] Y. Lei, N. Li, L. Guo, N. Li, T. Yan, and J. Lin, "Machinery health prognostics: A systematic review from data acquisition to RUL prediction," *Mechanical Systems and Signal Processing*, vol. 104, pp. 799–834, 2018.
- [2] L. Ren, Y. Liu, X. Wang, J. Lü, and M. J. Deen, "Cloud-edge-based lightweight temporal convolutional networks for remaining useful life prediction in IIoT," *IEEE Internet of Things Journal*, vol. 8, no. 16, pp. 12 578–12 587, 2021.
- [3] L. Basora, A. Viens, M. Arias Chao, and X. Olive, "A benchmark on uncertainty quantification for deep learning prognostics," *Reliability Engineering & System Safety*, vol. 253, p. 110513, 2025.
- [4] A. Javanmardi and E. Hüllermeier, "Conformal prediction intervals for remaining useful lifetime estimation," *International Journal of Prognostics and Health Management*, vol. 14, no. 2, 2023.
- [5] X. Xu, Y. Lu, B. Vogel-Heuser, and L. Wang, "Industry 4.0 and industry 5.0: Inception, conception and perception," *Journal of Manufacturing Systems*, vol. 61, pp. 530–535, 2021.
- [6] B. van Oudenhoven, P. Van de Calseide, R. Basten, and E. Demerouti, "Predictive maintenance for industry 5.0: behavioural inquiries from a work system perspective," *International Journal of Production Research*, vol. 61, pp. 7846–7865, 2023.
- [7] R. B. Saleh, K. P. Dinh, B. Villanyi, and T.-S. Hy, "HAMA: A hierarchical adaptive multi-agent architecture for industrial IoT predictive maintenance," *IEEE Access*, vol. 14, pp. 142 595–142 613, 2026.
- [8] L. Mendonça, J. Barbosa, and P. Leitão, "A human-centred architecture integrating digital twin and agentic AI for predictive maintenance," in *Proc. IEEE Int. Conf. Industrial Technology (ICIT)*, 2026, pp. 1–6.
- [9] K. A. Coulibaly and M. Hamlich, "Multi-agent system-driven digital twins for predictive maintenance: architectures, technologies and open research challenges," arXiv preprint arXiv:2607.21873, 2026.
- [10] Z. Yao, Y. Xu, T. Xia, Z. Chen, E. Pan, and L. Xi, "EC-Distill-ZeroDiag: a cloud-edge collaborative framework for zero-shot industrial fault diagnosis via large language model distillation," *Journal of Intelligent Manufacturing*, 2026.
- [11] Y. Romano, E. Patterson, and E. J. Candès, "Conformalized quantile regression," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 32, 2019.
- [12] R. F. Barber, E. J. Candès, A. Ramdas, and R. J. Tibshirani, "Predictive inference with the jackknife+," *The Annals of Statistics*, vol. 49, no. 1, 2021.
- [13] S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 30, 2017, pp. 4765–4774.
- [14] P. Nectoux, R. Gouriveau, K. Medjaher, E. Ramasso, B. Chebel-Morello, N. Zerhouni, and C. Varnier, "PRONOSTIA: An experimental platform for bearings accelerated degradation tests," in *Proc. IEEE Int. Conf. Prognostics and Health Management (PHM'12)*, Denver, CO, USA, 2012, pp. 1–8, hAL: hal-00719503.
- [15] B. Wang, Y. Lei, N. Li, and N. Li, "A hybrid prognostics approach for estimating remaining useful life of rolling element bearings," *IEEE Transactions on Reliability*, vol. 69, no. 1, pp. 401–412, 2020.
- [16] A. N. Angelopoulos, S. Bates, A. Fisch, L. Lei, and T. Schuster, "Conformal risk control," in *International Conference on Learning Representations (ICLR)*, 2024.